Luca **Tortorelli**

Astrophysicist & Science communicator



Contact

luca.tortorelli@physik.lmu.de
University Observatory Munich
LMU Munich
Scheinerstr. 1
81679 Munich, Germany
torluca.github.io
github.com/torluca
ORCID 0000-0002-8012-7495
ADS publication library
Geopop profile

Research focus

Galaxy evolution
Forward modelling
Simulation-based inference
Photometric redshifts
Stellar population synthesis
Wide-field surveys
Machine learning

Collaborations

4MOST (since 2022)
DESI (since 2022)
LSST DESC (since 2021)
DES (since 2021)
PAU Survey (since 2017)

Technical profile

Advanced: Python, astronomical data analysis, parallel/HPC workflows
Intermediate: PyTorch
Basic: C++, TensorFlow, Keras, SQL
HPC: multi-core jobs and SLURM

Profile

I develop physical and generative models that connect intrinsic galaxy populations to photometric and spectroscopic survey observations. My work combines forward modelling, simulation-based inference, stellar population synthesis and machine learning, with scientific leadership roles in 4MOST/WAVES and LSST DESC. I also write extensively for the public and am the author of the popular-science book *Galassie*.

Current appointment

10/2024–09/2027 **Akademischer Rat**
Fixed-term academic faculty, University Observatory Munich, Ludwig-Maximilians-Universität München; Prof. Dr. Daniel Grün research group, Munich, Germany.

Selected achievements

1. **GalSBI-SPS leadership.** Conceived and led the development, manuscript preparation and public release of the GalSBI-SPS galaxy population model. It replaces phenomenological descriptions with intrinsic physical properties propagated through spectral-energy-distribution and survey simulations. Applied to HSC, it reproduces key observed-property and redshift distributions and provides a reusable framework for Stage IV forward-modelling analyses.
2. **Survey-scale scientific leadership.** Co-lead the WAVES *SED Modelling and Galaxy Properties* Technical Working Group, coordinating more than 20 researchers developing methods to infer physical properties for approximately 1.7 million galaxies observed with 4MOST.
3. **Field-level synthesis.** Invited in 2026 to co-author the review *Machine Learning Techniques for Astrophysics and Cosmology: Photometric Redshifts*, recognising expertise spanning machine learning, generative galaxy modelling and redshift calibration.

Employment

03/2023–present **Freelance science writer**
Geopop, Milan, Italy.

10/2021–09/2024 **Postdoctoral researcher**
University Observatory Munich, LMU Munich, Germany.

09/2016–09/2021 **PhD researcher**
Institute for Particle Physics and Astrophysics, ETH Zurich, Switzerland.

09/2015–12/2016 **INAF Associate**
INAF Osservatorio Astronomico di Capodimonte, Naples, Italy.

09/2012–12/2012 **Laboratory technician and library assistant**
Department of Physics "Ettore Pancini", University of Naples Federico II, Italy.

Education

2016–2021 **PhD in Astrophysics, ETH Zurich**
Thesis: *The Measurement of Galaxy Population Properties with Forward-Modelling and Approximate Bayesian Computation*. Supervisors: Prof. Dr. Alexandre Refregier and Prof. Dr. Adam Amara.

2013–2016 **MSc in Physics, summa cum laude**
University of Naples Federico II. Thesis: *The Kormendy Relation of Galaxies in Clusters of the CLASH-VLT Survey*. Supervisors: Prof. Dr. Amata Mercurio and Prof. Maurizio Paolillo.

2009–2013 **BSc in Physics, summa cum laude**
University of Naples Federico II. Thesis: *Ellipticity Measurements in KiDS Survey Images Aimed at the Study of Gravitational Lensing*. Supervisors: Prof. Dr. Massimo Capaccioli and Prof. Dr. Nicola Napolitano.

Research programme and leadership

My scientific path has been shaped by transitions that added capabilities missing from the previous stage. During my PhD, I moved from individual galaxy properties to forward-modelling and simulation-based-inference methods connecting intrinsic populations to survey observations. At LMU, limitations of phenomenological modelling motivated my transition to physical galaxy population models, spectroscopy and machine-learning acceleration. As the programme matured, my role expanded from executing individual projects to defining scientific directions and coordinating teams. In LSST DESC, embedding generative modelling into the shared RAIL infrastructure required methods that are usable, testable and maintainable by a large collaboration. In 4MOST, I now contribute directly to the creation, validation and scientific interpretation of survey data products.

Survey and collaboration leadership

2026–present — **4MOST Infrastructure Working Group 6 member**
Quality assurance of 4MOST main-survey spectroscopic data.

2024–present — **Co-lead, WAVES SED Modelling and Galaxy Properties TWG**
Coordinate more than 20 researchers developing methods and products for physical-property inference from spectroscopy and multi-wavelength photometry.

2024–present — **Project Office member, 4MOST WAVES**
Wide Area VISTA Extragalactic Survey.

2024–present — **Collaboration Council member, LSST DESC**

2023–2025 — **Membership Committee member, LSST DESC**

2023 — **Postdoctoral representative**
University Observatory Munich.

Current — **4C3R2 Project Scientist**
Lead analysis of Survey Programme Validation data for very low signal-to-noise galaxies and the associated quality-control effort.

Scientific service and academic organisation

2025–present — **External expert proposal reviewer**
"Bando Giovani Ricercatori", Fondazione Cariplo, Italy.

2022–present — **Review Editor**
Extragalactic Astronomy, *Frontiers in Astronomy and Space Sciences*.

2021–present — **Scientific referee**
Monthly Notices of the Royal Astronomical Society and *Astrophysical Journal Letters*.

2024 — **Creator and main organiser**
2nd LMU–ETH Joint Workshop, ETH Zurich.

2023 — **Creator and main organiser**
1st LMU–ETH Joint Workshop, LMU Munich.

2021–present — **Seminar organiser**
Modern Statistics of Galaxies seminar series, LMU Munich.

Awards and third-party funding

04/2026 — **LMU–Todai Cooperation in the Sciences**
EUR 3,447 grant supporting a long research visit at Kavli IPMU, Tokyo.

01/2026 — **ORIGINS Visitor Programme**
Funding to host Dr. Alex Alarcon Gonzalez and Dr. Chris Lovell at LMU Munich.

06/2025 — **IAU Symposium 397 travel grant**
EUR 900 travel award for the UniversAI symposium in Athens.

Academic distinctions

Invited speaker at institute astrophysics colloquia and seminars of the Institute of Cosmology and Gravitation Portsmouth, ICRAR–University of Western Australia, Ruhr-Universität Bochum, Hamburg Observatory, Vienna University, ESO and Kavli IPMU.

Teaching and mentoring

University teaching

SS 2026	**Astrophysikalisches Praktikum** Designed and supervised the Galaxy Evolution experiment, LMU Munich.
WS 2025/26	**Observational Methods for Galaxy Surveys** Co-lecturer and exercise-class head, LMU Munich.
WS 2025/26	**Astrophysikalisches Praktikum** Designed and supervised the Galaxy Evolution experiment, LMU Munich.
SS 2025	**Astrophysikalisches Praktikum** Designed and supervised the Galaxy Evolution experiment, LMU Munich.
WS 2024/25	**Astrophysikalisches Praktikum** Designed and supervised the Galaxy Evolution experiment, LMU Munich.
SS 2024	**Astrophysikalisches Praktikum** Designed and supervised the Galaxy Evolution experiment, LMU Munich.
WS 2023/24	**Astrophysikalisches Praktikum** Designed and supervised the Galaxy Evolution experiment, LMU Munich.
SS 2023	**Observing and Data Analysis Methods for Cosmological Surveys** Co-lecturer and exercise-class head, LMU Munich.
WS 2022/23	**Astrophysikalisches Praktikum** Supervised the Interferometry experiment, LMU Munich.
SS 2022	**Observational Methods for Galaxy Surveys: From Pixels to Cosmology** Co-lecturer and exercise-class head, LMU Munich.
WS 2021/22	**Astrophysikalisches Praktikum** Supervised the Planetary Nebulae experiment, LMU Munich.
2021–present	**Modern Statistics of Galaxies** Adviser for BSc and MSc student presentations on current research, LMU Munich.
WS 2020	**Cosmological Probes** Co-lecturer and exercise-class head, ETH Zurich.
2016–2020	**Advanced Astrophysics Laboratory** Telescope operator, co-lecturer and teaching-material developer, ETH Zurich.

Student supervision

04/2026–present	**Elsa Dejby**, MSc, LMU Munich - mentor.
01/2026–present	**Rasila Banu**, MSc, University of Vienna - co-mentor.
10/2025–present	**Johanna Glueck**, MSc, LMU Munich - mentor.
09/2025–04/2026	**Julia Salustowicz**, MSc, ETH Zurich - co-mentor.
01/2025–02/2026	**Moritz Gammel**, MSc, LMU Munich - mentor.
09/2024–present	**Nikolas Frediani**, PhD, LMU Munich - co-supervisor.
06/2024–07/2025	**Jizhao Qin**, MSc, LMU Munich - mentor.
02/2024–07/2024	**Johanna Glueck**, BSc, LMU Munich - mentor.
02/2024–07/2024	**Elsa Dejby**, BSc, LMU Munich - mentor.
09/2023–present	**Zekang Zhang**, PhD, LMU Munich - co-supervisor.
04/2022–10/2022	**Paulina Contreras-Miranda**, BSc, LMU Munich - mentor.
10/2021–09/2022	**Benjamin Csizi**, MSc, LMU Munich - mentor.
09/2020–12/2020	**Michael Blanc**, BSc project, ETH Zurich - mentor.
03/2020–06/2020	**Pascal Hitz**, BSc project, ETH Zurich - mentor.
03/2019–09/2019	**Peter Strattman**, MSc, ETH Zurich - mentor.
03/2018–06/2018	**Beatrice Moser**, BSc project, ETH Zurich - mentor.

Teaching materials

- Designed and wrote the Astrophysical Lab manual for the experiment "Measuring the star-forming main sequence of COSMOS galaxies", LMU Munich.
- Designed and wrote the VP ASL-Astro HPP Telescope manual for the ETH Zurich 50 cm student telescope.
- Designed and wrote the data-reduction manual for the ETH Zurich 50 cm student telescope.

Invited talks and seminars

11/2026 3rd GWCats Workshop review talk, Toulouse, France (scheduled).

09/2026 ESO Lunch Talk, ESO, Germany.

06/2026 Simulation-based Inference in Galaxy Evolution 2026, Cambridge, UK.

04/2026 Astro Seminar, Kavli Institute for the Physics and Mathematics of the Universe, Tokyo, Japan.

03/2026 IAU Early Career Discourse Series, online.

11/2025 Vienna Astrophysics Colloquium, Vienna Observatory, Austria.

10/2025 Hamburger Sternwarte Colloquium, Hamburg Observatory, Germany.

11/2024 ICRAR–UWA Colloquium, International Centre for Radio Astronomy Research, Perth, Australia.

09/2024 2nd LMU–ETH Joint Workshop, ETH Zurich, Switzerland.

07/2024 Institute of Cosmology and Gravitation Seminar, Portsmouth, UK.

06/2024 OPINAS Seminar, Max Planck Institute for Extraterrestrial Physics, Garching, Germany.

09/2023 1st LMU–ETH Joint Workshop, LMU Munich, Germany.

Contributed talks and conference presentations

09/2026 Annual Meeting of the Astronomische Gesellschaft, Garching, Germany.

07/2026 LSST DESC Meeting, Astroparticule et Cosmologie, Paris, France.

07/2026 European Astronomical Society Meeting, Lausanne, Switzerland.

06/2026 WAVES Team Meeting, University of Sussex, Brighton, UK.

06/2026 4MOST All Hands Meeting, Leibniz Institute for Astrophysics Potsdam, Germany.

12/2025 Reconstructing the Universe, ASIAA, Taipei, Taiwan.

10/2025 Large Scale Structure Days, Garching, Germany.

07/2025 LSST DESC Meeting, online.

07/2025 New Strategies for Extracting Cosmology from Galaxy Surveys, Sexten, Italy.

07/2025 Tracing Cosmic Evolution with Galaxy Clusters V, Sexten, Italy.

06/2025 Cosmology from Home, online.

06/2025 Cosmological Surveys and Synergies (COLOURS), Institut Pascal, Orsay, France.

06/2025 UniversAI: Exploring the Universe with Artificial Intelligence, Athens, Greece.

05/2025 Simulation-based Inference in Galaxy Evolution, University of Bristol, UK.

05/2025 Big Data, Big Questions: The Future of Cosmological Surveys, MIAPbP, Garching, Germany.

12/2024 WAVES Team Meeting, University of Western Australia, Perth, Australia.

07/2024 DESI Meeting, online.

07/2024 LSST DESC Meeting, ETH Zurich, Switzerland.

Contributed talks and conference presentations (continued)

03/2024 MPE Meeting, Schloss Ringberg, Germany.

11/2023 Zooming PRIN Meeting, University of Milan, Italy.

07/2023 LSST DESC Meeting, SLAC, Stanford, USA.

06/2023 WE Heraeus–EAS Early Career Researchers in Astronomy, MPE, Garching, Germany.

02/2023 LSST DESC Meeting, online.

08/2022 LSST DESC Meeting, KICP, Chicago, USA.

07/2022 Zooming into Dark Matter and Proto-galaxies with Gravitational Lensing, Sexten, Italy.

11/2020 E-poster, 9th KIAS Workshop on Cosmology and Structure Formation, online.

02/2020 PAU Survey Collaboration Meeting.

07/2019 Tracing Cosmic Evolution with Clusters of Galaxies, Sexten, Italy.

07/2019 Swiss Cosmology Days, University of Zurich, Switzerland.

06/2019 Artificial Intelligence Methods in Cosmology, Centro Stefano Franscini, Switzerland.

03/2019 Cosmology on Safari, University of KwaZulu-Natal, South Africa.

09/2018 CLUSTER2 Meeting, University of Naples Federico II, Italy.

05/2018 PAU Survey Collaboration Meeting.

10/2017 PAU Survey Collaboration Meeting.

Observing experience

Observing runs

2026 — **DESI remote observing**
Four nights with the Dark Energy Spectroscopic Instrument on the Mayall Telescope, Kitt Peak, Arizona, USA.

2019 — **William Herschel Telescope**
Four nights with the PAU Camera, La Palma, Spain.

2018 — **William Herschel Telescope**
Three nights with the PAU Camera, La Palma, Spain.

2017–2020 — **VP ASL-Astro HPP Telescope**
Telescope operator and manual writer for the ETH Zurich student telescope.

Observing proposals

2020 — **ESO 0102.A-0640**
Co-I, "Probing the Nature of a Giant Gravitational Telescope and the Lensed Primordial Galaxies with MUSE", UT4-Yepun/MUSE, 12 h.

2019 — **ESO 0101.A-0470**
Co-I, "VST-GAME: Galaxy Assembly as a Function of Mass and Environment with VST", VST/OmegaCAM, 48 h.

2017 — **ESO 099.B-0242**
Co-I, "The MUSE Atlas of Disks", UT4-Yepun/MUSE, one night.

Research visits, schools and specialist training

04/2026 — **Kavli IPMU, Japan**
Established collaborations with IPMU cosmologists and developed a USpec plugin to simulate PFS spectrograph data.

11/2024 — **ICRAR, University of Western Australia**
4MOST collaboration and training in stellar population synthesis modelling.

01/2020 — **Deep Learning Meets (Astro)physics**
ETH Zurich; machine-learning training for astronomy.

10/2019 — **KIPAC, Stanford University**
Research visit and collaboration building with the Risa Wechsler group.

02/2019 — **Machine Learning for High Energy Physics**
University of Zurich; training in general machine-learning algorithms.

09/2015 — **Russian–Italian School on Astrophysics and Cosmology**
Dubna University, Russia.

09/2015 — **Astrophysical Probes of Fundamental Physics**
University of Ferrara, Italy; theoretical and hands-on astrophysical data analysis.

Science communication and outreach

Current — **Popular-science book author**
Author of *Galassie*, published by Edizioni La Serra and distributed in Italian bookstores.

03/2023–present — **Freelance popular-science writer**
More than 300 astronomy and space-science articles for Geopop, Milan, Italy.

09/2017 — Outreach experiments, ETH Scientifica, ETH Zurich.

05/2015 — Outreach experiments, Città della Scienza, Naples.

04/2015 — Outreach experiments, Astrokids, INAF Capodimonte, Naples.

10/2014 — Outreach experiments, MSC Lirica Cruise, Italy.

05/2014 — Outreach experiments, Città della Scienza, Naples.

09/2013 — Outreach experiments, European Researchers' Night, INAF Capodimonte, Naples.

05/2013 — Outreach experiments, Città della Scienza, Naples.

Refereed publications

1. Granata, **Tortorelli**, Grillo et al., "The velocity dispersion function of red galaxies in four Hubble Frontier Fields galaxy clusters", *A&A* 709, A254 (2026).
2. Tomasetti, Moresco, Granata et al., "Cosmic chronometers with galaxy clusters: A new avenue for multi-probe cosmology", *A&A* 708, A145 (2026).
3. Suyu, Acebron, Grillo et al., "Cosmology with supernova Encore in the strong lensing cluster MACS J0138-2155: Lens model comparison and H0 measurement", *A&A* 708, A291 (2026).
4. RAIL Team, van den Busch, Charles et al., "Redshift Assessment Infrastructure Layers (RAIL): Rubin-era photometric redshift stress-testing and at-scale production", *OJAp* 9, 58200 (2026).
5. Zhang, Grün, **Tortorelli** et al., "Emulating redshift mixing due to blending in weak gravitational lensing", *A&A* 706, A234 (2026).
6. **Tortorelli**, Fischbacher, Grün et al., "GalSBI-SPS: A stellar population synthesis-based galaxy population model for cosmology and galaxy evolution applications", *A&A* 703, A255 (2025).
7. Ertl, Suyu, Schuldt et al., "Cosmology with supernova Encore in the strong lensing cluster MACS J0138-2155: Photometry, cluster members, and lens mass model", *A&A* 702, A157 (2025).
8. D'Addona, Mercurio, Grillo et al., "Is there a tilt in the fundamental (hyper)plane?", *A&A* 700, A179 (2025).
9. Fischbacher, Kacprzak, **Tortorelli** et al., "GalSBI: phenomenological galaxy population model for cosmology using simulation-based inference", *JCAP* 2025, 007 (2025).
10. **Tortorelli**, McCullough and Grün, "Impact of stellar population synthesis choices on forward modelling-based redshift distribution estimates", *A&A* 689, A144 (2024).
11. Csizi, **Tortorelli**, Siudek et al., "The PAU Survey: Galaxy stellar population properties estimates with narrowband data", *A&A* 689, A37 (2024).
12. Khederlarian, Newman, Andrews et al., "Emission line predictions for mock galaxy catalogues: a new differentiable and empirical mapping from DESI", *MNRAS* 531, 1454 (2024).
13. D'Addona, Mercurio, Rosati et al., "The powerful lens galaxy cluster PLCK G287.0+32.9 ($\theta_E \sim 43''$): Redshift catalog and new lens model using MUSE observations", *A&A* 686, A4 (2024).
14. Manzoni, Baugh, Norberg et al., "The PAU Survey: a new constraint on galaxy formation models using the observed colour redshift relation", *MNRAS* 530, 1394 (2024).
15. Moser, Kacprzak, Fischbacher et al., "Simulation-based inference of deep fields: galaxy population model and redshift distributions", *JCAP* 2024, 049 (2024).
16. Berner, Refregier, Moser et al., "Fast forward modelling of galaxy spatial and statistical distributions", *JCAP* 2024, 023 (2024).
17. Granata, Bergamini, Grillo et al., "Exploring the low-mass regime of galaxy-scale strong lensing: Insights into the mass structure of cluster galaxies", *A&A* 679, A124 (2023).
18. Angora, Rosati, Meneghetti et al., "Searching for strong galaxy-scale lenses in galaxy clusters with deep networks. I. Methodology and network performance", *A&A* 676, A40 (2023).
19. **Tortorelli** and Mercurio, "morphofit: An automated galaxy structural parameters fitting package", *Frontiers in Astronomy and Space Sciences* 10, 51 (2023).
20. **Tortorelli**, Mercurio, Granata et al., "The Kormendy relation of early-type galaxies as a function of wavelength in Abell S1063, MACS J0416.1-2403, and MACS J1149.5+2223", *A&A* 671, L9 (2023).

Refereed publications (continued)

21 Meneghetti, Ragagnin, Borgani et al., "The probability of galaxy-galaxy strong lensing events in hydrodynamical simulations of galaxy clusters", *A&A* 668, A188 (2022).

22 Berner, Refregier, Sgier et al., "Rapid simulations of halo and subhalo clustering", *JCAP* 2022, 002 (2022).

23 Ragagnin, Meneghetti, Bassini et al., "Galaxies in the central regions of simulated galaxy clusters", *A&A* 665, A16 (2022).

24 Marques Oliveira, Sicardy, Gomes-Júnior et al., "Constraints on the structure and seasonal variations of Triton's atmosphere from the 5 October 2017 stellar occultation and previous observations", *A&A* 659, A136 (2022).

25 Granata, Mercurio, Grillo et al., "Improved strong lensing modelling of galaxy clusters using the Fundamental Plane: Detailed mapping of the baryonic and dark matter mass distribution of Abell S1063", *A&A* 659, A24 (2022).

26 **Tortorelli**, Siudek, Moser et al., "The PAU Survey: measurement of narrow-band galaxy properties with approximate Bayesian computation", *JCAP* 2021, 013 (2021).

27 Mercurio, Rosati, Biviano et al., "CLASH-VLT: Abell S1063. Cluster assembly history and spectroscopic catalogue", *A&A* 656, A147 (2021).

28 Johnston, Joachimi, Norberg et al., "The PAU Survey: Intrinsic alignments and clustering of narrow-band photometric galaxies", *A&A* 646, A147 (2021).

29 **Tortorelli**, Fagioli, Herbel et al., "Measurement of the B-band galaxy luminosity function with Approximate Bayesian Computation", *JCAP* 2020, 048 (2020).

30 Fagioli, **Tortorelli**, Herbel et al., "Spectro-imaging forward model of red and blue galaxies", *JCAP* 2020, 050 (2020).

31 Cabayol-Garcia, Eriksen, Alarcón et al., "The PAU Survey: background light estimation with deep learning techniques", *MNRAS* 491, 5392 (2020).

32 Eriksen, Alarcon, Gaztanaga et al., "The PAU Survey: early demonstration of photometric redshift performance in the COSMOS field", *MNRAS* 484, 4200 (2019).

33 Tonello, Tallada, Serrano et al., "The PAU Survey: Operation and orchestration of multi-band survey data", *Astronomy and Computing* 27, 171 (2019).

34 Cabayol, Sevilla-Noarbe, Fernández et al., "The PAU Survey: star-galaxy classification with multi narrow-band data", *MNRAS* 483, 529 (2019).

35 Stothert, Norberg, Baugh et al., "The PAU Survey: spectral features and galaxy clustering using simulated narrow-band photometry", *MNRAS* 481, 4221 (2018).

36 **Tortorelli**, Della Bruna, Herbel et al., "The PAU Survey: a forward modeling approach for narrow-band imaging", *JCAP* 2018, 035 (2018).

37 Fagioli, Riebartsch, Nicola et al., "Forward modeling of spectroscopic galaxy surveys: application to SDSS", *JCAP* 2018, 015 (2018).

38 **Tortorelli**, Mercurio, Paolillo et al., "The Kormendy relation of galaxies in the Frontier Fields clusters: Abell S1063 and MACS J1149.5+2223", *MNRAS* 477, 648 (2018).

Submitted and other relevant publications

1 Frediani, Grün, **Tortorelli** et al., "Data-driven Galaxy Population Prior for Photometric Redshifts", arXiv:2609.26594 (2026).

2 Zhang, Wang, Grün et al., "Anisotropic redshift distributions in photometric galaxy clustering and their cosmological impact", arXiv:2609.15542 (2026).

3 Fischbacher, **Tortorelli**, Kacprzak et al., "GalSBI: Forward Modelling Galaxy Clustering and Population", arXiv:2606.23783 (2026).

4 **Tortorelli** and Grün, "Machine Learning Techniques for Astrophysics and Cosmology: Photometric Redshifts", arXiv:2605.06790 (2026).

5 Contreras Miranda, **Tortorelli**, Grün et al., "Mapping the Zoo of Rest-Frame Galaxy SEDs using unsupervised machine learning", *RMxAC* 60, 252 (2026).

6 **Tortorelli**, Fischbacher, Robotham et al., "ProMage: fast galaxy magnitudes emulation combining SED forward-modelling and machine learning", arXiv:2509.00150 (2025).

7 Fischbacher, Moser, Kacprzak et al., "galsbi: A Python package for the GalSBI galaxy population model", arXiv:2412.08722 (2024).

8 Fischbacher, Moser, Kacprzak et al., "UFig v1: The ultra-fast image generator", arXiv:2412.08716 (2024).

9 Angora, Rosati, Meneghetti et al., "Simulating high-realistic galaxy scale strong lensing in galaxy clusters to train deep learning methods", *IAUS* 381, 85 (2024).

10 Grün, McCullough, Amon et al., "4MOST Complete Calibration of the Colour-Redshift Relation (4C3R2)", *The Messenger* 190, 28 (2023).

11 Meneghetti, Ragagnin, Borgani et al., "Too many galaxy-galaxy strong lenses observed in galaxy clusters", *MmSAI* 94, 67 (2023).

12 Granata, Mercurio, Grillo et al., "Investigating the discrepancy in sub-halo compactness between observed and simulated galaxy clusters with improved strong lensing modelling", *MmSAI* 94, 63 (2023).

Luca Tortorelli • luca.tortorelli@physik.lmu.de • torluca.github.io